

POWER · STACK · PLANT · MOLECULE · MONEY

Where the Money Is

A map of the green hydrogen value chain, India first, to find where a new entrant can build a durable, profitable business by 2035.

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COMPANION TO

Dossier 1: *Without the Hype*

PRM project with Shalvi Bhatnagar

Prof. P. R. C. Gopal

Every figure is tagged by what kind of number it is: REPORTED *a company or government actually published it;* ESTIMATED *my calculation or a reasoned estimate;* PROJECTED *a target or forecast that has not happened yet. Claims that matter also carry a confidence tag (*HIGH*,* MEDIUM*,* LOW*).*

THE ANSWER FIRST

The money is not where the hype is

What I found, in one page, before the evidence.

I went looking for the best business in green hydrogen and found something uncomfortable. **The layers everyone talks about, electrolysers and giant export plants, are where money is being lost.** The layers nobody talks about are where it is being made.

- **Owning contracted power and molecules pays.** A renewable power producer like ACME Solar runs at a 90.5% EBITDA margin [?] REPORTED. Industrial-gas companies that sell hydrogen as a service earn about 30% operating margins [?] REPORTED. Both sell long contracts.
- **Making the machines does not pay, yet.** thyssenkrupp nucera, a major electrolyser maker, made about €2 million of operating profit on €845 million of sales [?] REPORTED. Enapter lost €18.1 million of EBITDA on €22.1 million of revenue, partly from returned stacks [?] REPORTED. Plug Power hit its first positive gross margin in years only in late 2025, after a going-concern warning and a 135% rise in share count [?, ?] REPORTED.
- **The building in between barely pays.** India's largest solar EPC contractor earned about a 4.6% EBITDA margin in FY25 [?] ESTIMATED.

Economists call this shape a *smile curve*: profits pool at the two ends of the chain (owning the energy, owning the customer) and drain out of the middle (making and building). For a new entrant with little capital, that points away from factories and mega-plants and towards **services and contracts that sit close to the customer**: hydrogen-as-a-service for mid-sized factories, independent servicing of electrolyser stacks, and green-certification data. Section ?? ranks the five best entry points.

> *"In a gold rush, the people who got rich sold shovels. In green hydrogen, even the shovel-makers are losing money. The people who will get rich will be the ones who keep the mines running."*

HOW THIS CONNECTS TO DOSSIER 1

Dossier 1 asked which machine our plant should use. This one asks a bigger question: *if we were building a company, not a plant, where in the chain should it stand?* The two answers fit together. The plant is the project for Prof. Gopal; the business that services, certifies and supplies plants like it is the long-term venture.

SECTION ONE

The value chain, layer by layer

Fifteen layers from sunlight to finance: who makes money in each, how much it costs to play, and how hard it is to get in.

1.1 The map

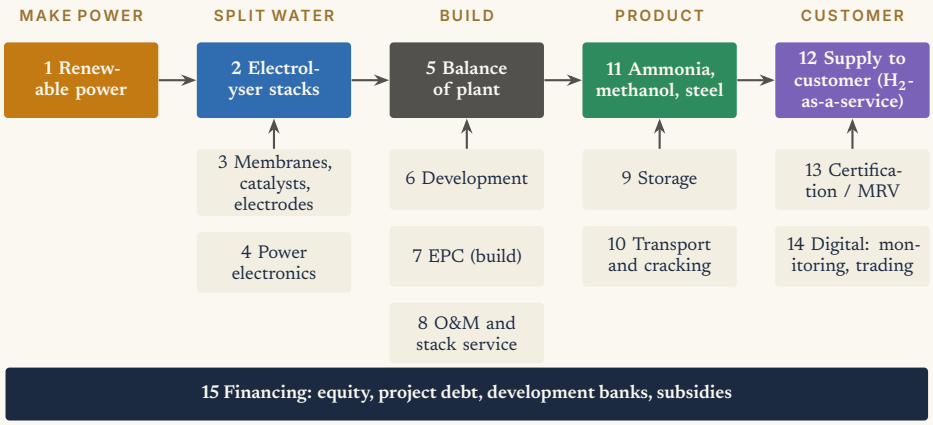


Figure 1. *The green hydrogen value chain. The top row is the physical flow; the boxes below each are the businesses that serve that step. Financing runs under everything.*

1.2 The economics of each layer

Scores use the five-step scale from Dossier 1. Here, a 5 under *capital intensity*, *concentration* or *barriers* means that column is high (usually bad for a newcomer); a 5 under *growth* means fast growth to 2035 (good).

Table 1. *Value-chain economics. Margins are typical ranges with the anchor evidence named; scores are my judgement.*

LAYER	TYPICAL MARGIN (EVIDENCE)	CAPITAL	CONCEN.	BARRIERS	GROWTH
1 Renewable power	EBITDA 85–90%: ACME Q4 FY25 90.5% [?] REPORTED; returns on capital far lower because assets are huge ESTIMATED	5	3	4	5
2 Electrolyser stacks	Around zero or negative: nucera EBIT about €2m on €845m [?]; Enapter EBITDA −€18.1m on €22.1m [?] REPORTED	4	5	4	2
3 Membranes, catalysts, electrodes	20–40% gross for incumbents with IP ESTIMATED	3	5	5	3
4 Power electronics	10–15% EBITDA, like other industrial electrical kit ESTIMATED	3	4	3	3
5 Balance of plant (water, compression, gas clean-up)	8–15% EBITDA ESTIMATED	2	2	2	3

LAYER	TYPICAL MARGIN (EVIDENCE)	CAPITAL	CONCEN.	BARRIERS	GROWTH
6 Project development	Lumpy: earned as a premium when a built, contracted plant is sold or financed ESTIMATED	3	3	3	3
7 EPC (build)	4–10% EBITDA: Sterling & Wilson FY25 about 4.6% [?] ESTIMATED	2	3	3	3
8 O&M and stack service	15–25% EBITDA, typical of industrial service contracts ESTIMATED	1	4	2	4
9 Storage	Utility-like regulated returns, once demand exists ESTIMATED	5	2	4	2
10 Transport and cracking	Pipelines utility-like; shipping cyclical; cracking pre-commercial ESTIMATED	5	3	5	2
11 Ammonia, methanol, green steel	Thin: SECI green ammonia averaged ₹53.27/kg, about 19% above grey [?] ESTIMATED	5	3	4	4
12 Supply to customer (H_2-as-a-service)	About 30%: Linde adjusted operating margin 29.8% (FY2025) [?] REPORTED	3	4	3	4
13 Certification / MRV	20–40% for verification services; software 60–80% gross ESTIMATED	1	3	3	4
14 Digital: monitoring, maintenance, trading	Software economics, 60–80% gross once at scale ESTIMATED	1	2	2	3
15 Financing	Lenders earn a 2–3% spread; equity earns the residual ESTIMATED	5	4	4	4

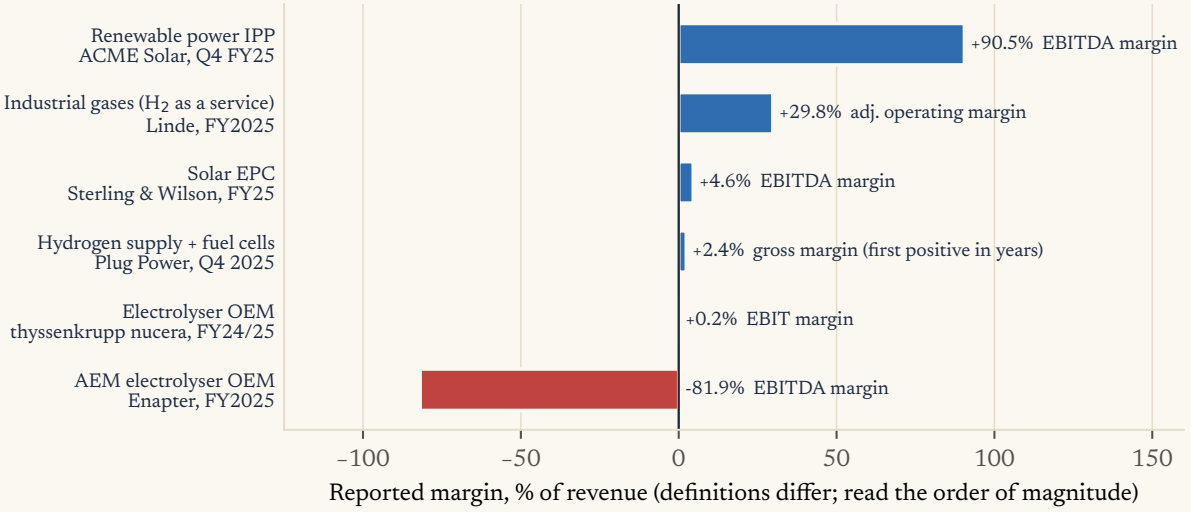


Figure 2. *Profit pools in practice: one reported margin from each type of business. Sources: [?, ?, ?, ?, ?, ?]. The EPC figure is my division of reported EBITDA by revenue.*

1.3 Who plays where

Lists are illustrative, not exhaustive. Where I name a company without a citation, I am naming its line of business, not making a claim about its numbers.

Table 2. *Key players and growth signals by layer.*

LAYER	INDIA	GLOBAL	GROWTH SIGNAL (DATED)
1 Power	ACME, Greenko, ReNew, Avaada, NTPC Renewable Energy	ACWA Power, Masdar, Iberdrola	Solar plus 4-hour storage at ₹2.86/kWh (Oct 2025) [?] REPORTED
2 Stacks	Incentive winners: Reliance 300 MW, John Cockerill Greenko 300, Jindal India 300, Adani 198.5, Ohmium 137, Homihydrogen 101.5, Advait 100, L&T 63 [?] REPORTED	Peric, Longi, Sungrow (China); thyssenkrupp nucera, Nel, ITM, Siemens Energy; Electric Hydrogen	58 GW/yr of factories ran at about 9% in 2025 [?] REPORTED
3 Components	Newtrace (alkaline electrodes) [?]	Johnson Matthey, De Nora, Heraeus, Chemours	ITM cut iridium a further 40% (Nov 2024) [?] REPORTED
4 Power electronics	Domestic power-equipment makers	ABB, Siemens, Hitachi Energy, Sungrow	Grows with every MW of electrolysis ESTIMATED
5 Balance of plant	Thermax, Ion Exchange	Atlas Copco, Burckhardt Compression	Desalination adds at most US$0.10/kg [?] REPORTED
6 Development	ACME, AM Green, Avaada, NTPC, Hygenco	ACWA Power, Air Products	862,000 t/yr allocated, about 8,000 t/yr built [?, ?] REPORTED
7 EPC	L&T, Tata Projects, Sterling & Wilson	Technip Energies, Worley, Toyo	AM Green says all major contracts awarded (Jan 2026) [?] REPORTED
8 O&M, stacks	Mostly OEM service agreements; independents scarce	OEMs; industrial service firms	OEM distress: Cummins stopped selling electrolysers [?]; Enapter stack returns [?] REPORTED
9 Storage	No large hydrogen storage; buffer tanks at plants	Salt caverns (US, Europe)	Depends on hydrogen trade emerging ESTIMATED
10 Transport	Port hubs at Kandla, Paradip, Tuticorin [?]	Ammonia carriers; cracking: Topsoe, Amogy [?]	Exporters ship ammonia, so cracking happens at the importer's end ESTIMATED
11 Derivatives	AM Green, ACME, NTPC REL, Jakson, Onix, SCC, Oriana [?]; 5 steel pilots [?]	Yara, CF Industries	Uniper to take up to 500,000 t/yr from AM Green from 2028 [?] PROJECTED
12 H_2-as-a-service	Hygenco (Jindal Stainless, BOOT) [?]; Linde India, INOX Air Products	Linde, Air Liquide, Air Products	Industrial-gas margins near 30% [?] REPORTED
13 Certification	BEE and accredited verification agencies under the 2025 certification scheme [?]	DNV, TÜV, Bureau Veritas	EU buyers want RFNBO-certified ammonia [?] REPORTED
14 Digital	Nascent; few hydrogen-specific players found	Siemens, AVEVA, Honeywell	Grows with plant count and certification rules ESTIMATED
15 Financing	REC (₹4,000 crore loan for ACME's Oman project) [?]; IREDA, PFC, SBI	World Bank (US$1.5 bn for India's low-carbon transition) [?]; GIC, ADIA [?]	Large balance sheets dominate; small tickets underserved ESTIMATED

1.4 Reading the map: margin against capital

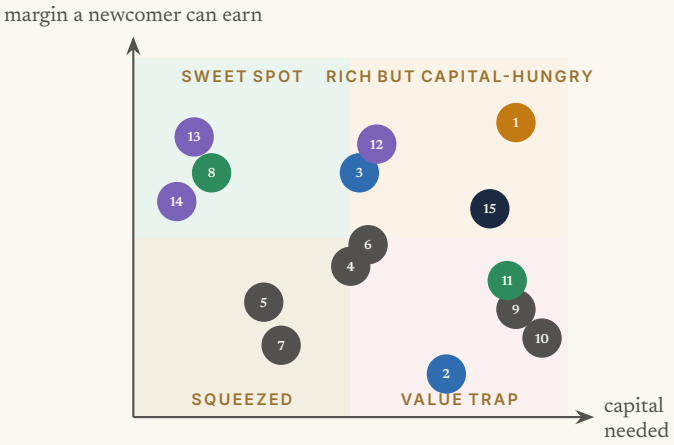


Figure 3. *My placement of the 15 layers (numbers match Table 1). Upper-left is where a capital-light newcomer can earn good margins.*

KEY FINDING • A SMILE CURVE, WITH A SWEET SPOT NEAR THE CUSTOMER

The highest margins sit at the two ends: owning contracted power or contracted supply (layers 1, 12) and owning knowledge (layers 3, 13). Both ends are capital-hungry *except* the knowledge end. The middle, making stacks and building plants (layers 2, 5, 7), is squeezed by Chinese overcapacity and reverse auctions. A newcomer without a balance sheet should aim for the upper-left: **O&M and stack service (8), certification and MRV (13), digital (14), and a capital-light version of hydrogen-as-a-service (12).** MEDIUM

SECTION TWO

Six ways to make money

The same molecule supports very different businesses. They differ less in technology than in how much capital they lock up and how long they wait to be paid.

2.1 The six models side by side

Table 3. *Business models compared. Capital figures are orders of magnitude for an entrant's first serious step.*

MODEL	CAPITAL NEEDED	TIME TO PROFIT	RISK PROFILE	EXIT ROUTES	REAL EXAMPLES
Developer / IPP (each plant in its own SPV, 70–80% debt)	₹2,900–3,600 crore for a 100 MW green-ammonia plant with its own renewables ESTIMATED	5–8 years: 2–3 to build, then contract cash flows	Construction, offtake and winner's-curse risk; returns fixed by contract	Sell operating SPVs to infra funds; IPO (ACME raised ₹2,900 crore in 2024 [?] REPORTED)	ACME, AM Green, Avaada, NTPC REL
Technology / IP licensing	US$10–100 m+ of R&D before revenue ESTIMATED	7–10 years	Technology risk first, then adoption risk	Licence fees; acquisition by an OEM	Versogen licensing AEM to an Indian factory [?]; Newtrace selling electrodes into others' stacks [?]
Equipment manufacturing	US$100–400 m to reach GW scale: EH2 raised US$380 m in one round [?]; Ohmium US$250 m [?] REPORTED	5–10 years; many never get there	Overcapacity, price war with China, warranty claims	IPO or trade sale; distressed sale is common	Ohmium, Electric Hydrogen, Enapter, Plug Power
Hydrogen-as-a-service (build, own, operate on the customer's site)	₹10–100 crore per site for 0.5–5 MW ESTIMATED	3–5 years per site	Customer credit; utilisation; technology at small scale	Sell the portfolio to an industrial-gas major or infra fund	Hygenco for Jindal Stainless [?]; Linde and INOX Air Products in grey hydrogen
Asset-light software / MRV	₹5–50 crore ESTIMATED	2–4 years	Small market until rules bite; regulation can help or hurt	Acquisition by a certifier, industrial-software or carbon-market firm	Accredited verification agencies under India's 2025 scheme [?]; few dedicated Indian platforms yet

MODEL	CAPITAL NEEDED	TIME TO PROFIT	RISK PROFILE	EXIT ROUTES	REAL EXAMPLES
Aggregator / trader	Mostly working capital and guarantees ESTIMATED	Fast if a state backs it; slow otherwise	Price and counterparty risk sits with the aggregator	Rarely sold; usually state-backed	SECI's 13 green-ammonia auctions [?]; Germany's H2Global, which cleared at US$1,153/t in 2024 [?] REPORTED

2.2 Scoring the models for a new Indian entrant

Table 4. *Each model scored 1–5, where 5 is better for a capital-light newcomer.*

MODEL	LOW CAPITAL	FAST CASH	LOW RISK	MARGIN	DEFENSIBLE	AVERAGE
Developer / IPP	1	2	3	4	3	2.6
Technology / IP licensing	2	1	1	5	5	2.8
Equipment manufacturing	1	1	1	2	2	1.4
Hydrogen-as-a-service	3	3	3	4	3	**3.2**
Asset-light software / MRV	5	4	3	4	2	**3.6**
Aggregator / trader	3	2	2	2	2	2.2

KEY FINDING • START LIGHT, EARN THE RIGHT TO GET HEAVY

No single model is both easy to start and defensible. Software and MRV are the cheapest to start but the easiest to copy. Hydrogen-as-a-service makes real margins but needs project money per site. IP is the most defensible and the slowest. The way through is a *sequence*: start where capital is lightest (data and services), use the customer relationships and operating data to win hydrogen-as-a-service contracts, and only then take on developer-scale balance-sheet risk. That sequence is also the outline of the long-term venture blueprint. MEDIUM

WHY EQUIPMENT MANUFACTURING SCORES LOWEST

It is the layer every government is subsidising, including ours, which is exactly why it is crowded. Global factories could make about 58 GW a year but made under 5 GW in 2025 [?]. When supply is twelve times demand, prices fall below cost and the weakest makers fail. A newcomer would be entering the most crowded room at the worst time.

SECTION THREE

How ten companies grew, or did not

Four Indian developers, two Indian technology start-ups, three global technology companies and one cautionary tale.

3.1 The Indian developers: power first, molecules second

Table 5. *Indian developers. All figures* REPORTED *unless tagged otherwise.*

COMPANY	FOUNDING AND CAPITAL	FIRST HYDROGEN STEP AND INFLECTION	WHERE IT STANDS (2025–26)	MISTAKE OR LESSON
ACME	Founded by Manoj Upadhyay [?]; ACME Solar Holdings incorporated 2015; IPO of ₹2,900 crore in Nov 2024 (₹2,395 crore fresh) [?]	Built what may be the world's first green-ammonia plant at Bikaner in 2021 [?]. Inflection: won 370,000 t/yr, 51% of SECI's Tranche I [?]	FY25 revenue ₹1,575 crore, profit ₹251 crore, 2,705 MW operating, 6,970 MW portfolio [?]. Gopalpur: 0.4 Mt/yr to Japan's IHI, first output likely 2027 [?] PROJECTED	Built a cash-generating power business first, then used it as the platform. Risk: half the national auction volume rides on one bidder's cost assumptions.
Greenko / AM Green	Founder Anil Chalamalasetty [?]. GIC (majority) and ADIA added US$329 m in 2019, part of US$824 m committed [?]. In 2025 AM Green bought ORIX's 17.5% of Greenko, taking its group to 25% [?]	FID in 2024 on a 1 Mt/yr green-ammonia plant at Kakinada, firmed by 950 MW of pumped storage, with John Cockerill alkaline electrolysers [?]. Inflection: Uniper signs for up to 500,000 t/yr (Jan 2026) [?]	Over 10 GW of renewables and pumped storage [?]. First shipment to Uniper "as early as 2028" [?] PROJECTED	Pumped storage gives it the steadiest power in India, a real moat. But production was guided to the second half of 2026 at FID [?]; shipments now start in 2028. Even the best-capitalised project slipped.
ReNew	Listed in New York; Canada's CPPIB holds 31.3% and ADIA 23.8% [?]	No recent hydrogen milestone that I could verify; I treat its hydrogen plans as dormant LOW	10.3 GW installed. A take-private bid at US$8.15 a share collapsed when Masdar withdrew (Dec 2025); shares fell about 28% to US$5.26 [?]	Size does not guarantee a hydrogen business. Capital structure and investor patience matter as much as assets.
Avaada	Founded by Vineet Mittal; raised US$1.07 bn in April 2023, led by Brookfield's transition fund, explicitly for green hydrogen and ammonia [?]	Bid in five SECI green-ammonia auctions and came second or third in all of them, often by ₹0.25/kg [?]	About 4 GW of renewables at the time of the raise, targeting 11 GW by 2026 [?] PROJECTED	Raising money early is not the same as winning. Avaada lost every auction to a slightly lower bid, which is either discipline or a cost gap. Either way, capital alone is not an edge.

3.2 The technology companies: India and the world

Table 6. *Technology start-ups and manufacturers. All figures* REPORTED *unless tagged otherwise.*

COMPANY	FOUNDING AND CAPITAL	PRODUCT AND INFLECTION	WHERE IT STANDS (2025–26)	MISTAKE OR LESSON
Ohmium (US HQ, India factory)	US$45 m Series B led by Fenice [?]; US$250 m Series C led by TPG Rise Climate, April 2023 [?]	PEM electrolysers built in Bengaluru, scaling from 500 MW to 2 GW a year [?] PROJECTED; won 137 MW of India's manufacturing incentive [?]	I could not verify recent orders or financial results LOW	Raised heavily into a market that then shrank. Watch whether its factory runs near capacity.
Newtrace (Bengaluru)	Founded 2021 by Prasanta Sarkar and Rochan Sinha; US$6.3 m pre-Series A in March 2026, led by HDFC Bank and Mitsui Sumitomo Insurance VC [?]	Sells electrodes ("Voltagen") into other makers' alkaline stacks rather than building whole systems [?]	Pilot manufacturing; no named customers yet [?]	A deliberately capital-light "picks and shovels" model: it wins if *any* alkaline maker wins.
Electric Hydrogen (US)	US$380 m Series C (Oct 2023), the first hydrogen-technology company valued at US$1 bn [?]	Sells whole 100 MW plants (about 50 t of hydrogen a day), not stacks; Fortescue framework for 1 GW; claims an order book above 5 GW [?]	In 2025 partnered with Titan for modular, factory-built plants [?]	Attacked the real cost problem, the balance of plant, rather than the stack alone. Order-book claims still need converting into delivered plants.
Hysata (Australia)	US$111.3 m Series B (May 2024), co-led by bp Ventures and Templewater, plus POSCO and others [?]	Capillary-fed electrolyser claiming efficiency "well above" IRENA's 2050 target; no number published in the release [?]	Scaling manufacturing towards gigawatt scale [?] PROJECTED	A genuine efficiency bet. Until independent kWh/kg data is public, treat the claim as unproven LOW.
Enapter (Germany, AEM)	Listed company	Pioneer of small, modular AEM electrolysers	2025 revenue €22.1 m, EBITDA −€18.1 m, including €4.5 m for *returned stacks* [?]	The clearest evidence that AEM durability is not yet solved. Directly relevant to Dossier 1's AEM option.

3.3 The cautionary tale: Plug Power

Plug Power is the case every hydrogen founder should study, because it did almost everything the market rewards in a boom and paid for all of it in the bust.

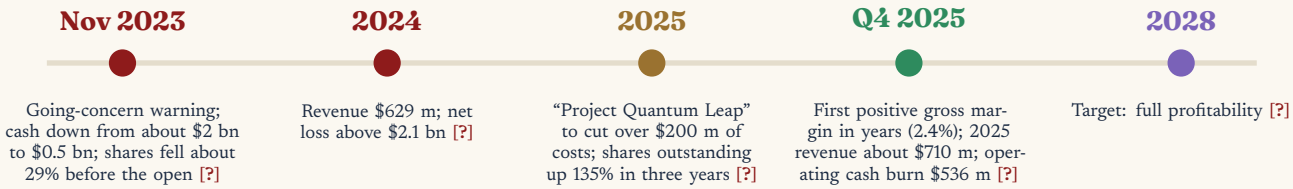


Figure 4. *Plug Power, 2023–2028. All figures* REPORTED*; the 2028 target is* PROJECTED.

What went wrong, in plain words. Plug sold equipment and hydrogen to win volume before it could make either at a profit; the company itself blamed "unprecedented hydrogen supply challenges" when its 2023 sales disappointed [?]. It funded the gap by selling new shares, so the share count rose 135% in three years and the stock lost about 85% [?]. It is now cutting costs and selling assets to reach profit by 2028 [?]. HIGH

FIVE RULES I TAKE FROM THE TEN CASES

1. **Cash flow first, hydrogen second.** ACME and Greenko built profitable power businesses before betting on molecules. The pure hydrogen plays (Plug, Enapter) burned cash.
2. **Never sell below cost to buy growth.** Plug's volume came with negative margins and a near-death experience.
3. **Contracts beat capital.** Avaada raised US$1 bn and still lost every auction. AM Green's turning point was a signed offtake, not a funding round.
4. **Durability is the hidden cost of new technology.** Enapter's returned stacks are a real charge on real accounts.
5. **Even the best slip.** AM Green's first exports moved from 2026 to 2028. Plan for delay as the base case, not the risk case.

MEDIUM

SECTION FOUR

The less-chosen technologies

Seven bets that most of the market is not making yet. For each: how ready it is, which way costs are heading, who is funding it, and what has to happen for it to win.

4.1 How to read readiness

Technology readiness level (TRL) runs from 1 (an idea on paper) to 9 (proven in commercial operation). The TRLs below are **my estimates**, based on the most advanced public deployment I could find, so every one is ESTIMATED.

Table 7. *Seven technology bets.*

BET	TRL	EVIDENCE AND COST DIRECTION	WHO IS BACKING IT	WHAT MUST HAPPEN TO WIN	FIT FOR AN INDIAN ENTRANT
AEM electrolysis	6–7	First 5 MW system delivered (April 2026) [?]. Enapter took €4.5 m of returned-stack costs in 2025 [?] REPORTED. It uses nickel catalysts rather than precious metals [?], so costs should fall with scale ESTIMATED	Enapter, Versogen (licensing a 250–300 MW Indian factory) [?]	20,000+ hours of field durability at multi-MW scale; makers that survive long enough to honour warranties	●●●○○ Good as a supply-chain play; risky as a plant foundation
SOEC	6–7	IEA capex range US$2,800–5,600/kW [?]. Leader Topsoe "hibernated" its factory and cancelled a US plant (June 2026) [?] REPORTED	Topsoe, Sunfire, Bloom Energy	Stack life far beyond 20,000 h; cheap industrial waste heat next door	●○○○○ Wrong fit for solar-led India today
Low-iridium PEM	8–9	ITM cut iridium a further 40% (Nov 2024) on top of earlier cuts over 80% [?]. World iridium supply is only 7–8 t a year [?] REPORTED	ITM, Johnson Matthey, Electric Hydrogen	Thrifting plus 90%+ recycling of iridium at end of life [?]	●●●○○ Catalyst recycling is an Indian niche worth a look

BET	TRL	EVIDENCE AND COST DIRECTION	WHO IS BACKING IT	WHAT MUST HAPPEN TO WIN	FIT FOR AN INDIAN ENTRANT
Direct seawater electrolysis	4–5	A membrane-based seawater electrolyser ran for over 3,200 hours in lab conditions [?] REPORTED MEDIUM. But desalinating water first costs at most US$0.10/kg of hydrogen [?]	Chinese academic and state groups	A cost saving larger than US$0.10/kg, which is hard; mainly useful offshore	●○○○○ Solves a problem that cheap desalination already solves
Ammonia cracking	6–7 at scale	Amogy: nearly US$300 m raised, a US$700 m valuation, an ammonia tugboat in 2024, a 40 MW power plant targeted for 2028–29 [?] REPORTED PROJECTED	Amogy, Topsoe, KBR	Large import terminals in Europe and Japan	●●○○○ India exports ammonia; cracking happens at the buyer's end
Natural (white) hydrogen	3–5	Koloma raised US$245 m (Series B) to find underground hydrogen; deposits so far found mostly by accident, as in Mali [?] REPORTED. The IEA warns deposits may be too scattered to pay [?]	Koloma (Khosla, Breakthrough Energy Ventures)	Proof of large, repeatable, cheap-to-tap reservoirs	●○○○○ Exploration capital and geology data an MBA start-up lacks
Biomass and waste gasification	5–7	Counts as "green" under India's standard [?]. NTPC NETRA has proposed a 1 t/day plasma-gasification plant on municipal and farm waste (Dec 2025) [?] PROJECTED	NTPC, Indian research institutes	Cheap, reliable feedstock collection; meeting the 2 kg CO_2 limit with certification	●●○○○ Interesting for small, distributed plants near waste sources

4.2 Patents and money: what the funding trail says

The best study of hydrogen patents, by the European Patent Office and the IEA, covers 2011–2020. The EU held 28% of international hydrogen patent families, Japan 24% and the US 20%. Electrolysis drove the growth in clean-production patents. The finding that matters most for a founder is about money: **start-ups that held patents received over half of the roughly US$10 bn of venture capital invested in hydrogen firms, and over 80% of the late-stage money**, even though they were under a third of the start-ups [?] REPORTED. HIGH

KEY FINDING • IN A TECHNOLOGY BET, THE PATENT IS THE TICKET

If we ever make a technology bet, the investment case starts with defensible IP, not with a pilot plant. That makes the realistic Indian technology plays *component* bets (electrodes, catalysts, catalyst recycling, power electronics tuned for variable solar) rather than whole-stack bets. Newtrace has already chosen this route [?]. MEDIUM

MY RANKING OF THE SEVEN BETS FOR US

AEM and **low-iridium PEM with catalyst recycling** are the only two worth tracking as a business, and both through components and services, not plants. Seawater electrolysis and ammonia cracking solve problems India does not have. SOEC and natural hydrogen need capital, heat or geology a start-up cannot reach. Biomass is a small, local niche: worth a pilot, not a strategy. MEDIUM

SECTION FIVE

What India is missing

Specific needs in the Indian market that nobody serves well yet, each with the evidence that it is real.

Table 8. *Unmet needs in the Indian green hydrogen market. "Size" is how big the need is; "Fit" is how well a capital-light start-up could serve it (both 1–5).*

GAP	EVIDENCE IT IS REAL	WHO WOULD PAY	SIZE	FIT
1. Building what has been allocated	862,000 t/yr allocated but about 8,000 t/yr commissioned by Feb 2026 [?, ?] REPORTED; AM Green's first exports moved to 2028 [?]	Developers, lenders	5	2
2. Stack servicing and independent testing	Eight Indian stack makers won incentives, some at only 100 MW a year [?]; global OEMs are under strain (Cummins exit [?], Enapter returns [?]); only 5 testing facilities approved nationally [?] REPORTED. A plant whose OEM fails has nobody to service it	Plant owners, lenders, insurers	4	4
3. Certification and data	India's certification scheme launched in 2025, run by BEE through accredited verifiers [?]; EU buyers such as Uniper want RFNBO-certified ammonia [?]; the EU border carbon charge covers hydrogen and ammonia [?]. Exporters need to satisfy two rulebooks with hourly data	Producers, exporters	3	5
4. Mid-sized industrial users	Glass, electronics, chemicals and chlor-alkali users are many and small [?]; Jindal Stainless already buys green hydrogen from an on-site plant built, owned and operated by Hygenco [?] REPORTED	Steel, glass, food, pharma, electronics plants	3	4
5. Electrolyser-ready water	20–30 L of water per kg once purification is counted [?]; port hubs at Kandla, Paradip and Tuticorin are coastal [?]	Developers, EPCs	3	3
6. Demand certainty	Refinery tenders were slow to arrive in 2025 [?]; purchase mandates are discussed but not notified [?]	The state	5	1
7. Small-ticket finance	Big projects draw on REC (₹4,000 crore to ACME for Oman) [?] and the World Bank [?]; nothing comparable is visible for ₹10–100 crore on-site plants ESTIMATED	Banks, NBFCs, insurers	3	2

KEY FINDING • THE BIGGEST GAPS ARE NOT THE ONES A START-UP SHOULD CHASE

Gaps 1 and 6, building the allocated plants and creating demand, are the largest, but they need state power or billions of rupees. Gaps 2, 3 and 4 are smaller but match a capital-light, analytics-heavy team. They also reinforce

each other. A company that services stacks collects the operating data; that data feeds certification; and certified, well-serviced plants are what a hydrogen-as-a-service contract sells. MEDIUM

AN HONEST LIMIT OF THIS SECTION

The evidence for gaps 4 and 7 is thinner than for the others. I found one strong example of on-site green hydrogen for a mid-sized user (Jindal and Hygenco) and no good public data on what small Indian users pay for cylinder hydrogen today. Before building a business on gap 4, we would need to interview a dozen glass, food and electronics plants about their volumes and prices. That is a perfect field exercise for the research paper.

SECTION SIX

Five entry points, ranked

Where a new entrant can build a durable, profitable business by 2035, judged on how attractive it is and how feasible it is for a capital-light team.

6.1 How I scored them

Attractiveness is the average of three 1–5 scores: margin, growth to 2035 and defensibility. **Feasibility** is the average of capital fit, capability fit (for a team with finance, analytics and engineering skills) and speed to first revenue. The ranking uses attractiveness × feasibility, so an idea must be good on both axes to rank high. All scores are my judgement ESTIMATED, built on the evidence in Sections 1–5.

Table 9. *The ranked entry points.*

RANK	ENTRY POINT	MARGIN	GROWTH	DEFEND	ATTRACT.	CAPITAL	SKILLS	SPEED	FEASIB.
1	Independent stack service, testing and refurbishment	4	4	4	**4.0**	4	4	4	**4.0**
2	Certification and hourly MRV platform	4	4	2	**3.3**	5	5	4	**4.7**
3	Hydrogen-as-a-service for mid-sized factories	4	4	3	**3.7**	2	3	3	**2.7**
4	Electrolyser-ready water and balance-of-plant packages	3	3	2	**2.7**	3	3	4	**3.3**
5	Component IP: electrodes, catalysts, iridium recycling	5	3	5	**4.3**	2	2	1	**1.7**
–	*Not ranked:* developer / IPP at scale	4	4	1	3.0	1	1	1	1.0
–	*Not ranked:* stack manufacturing	1	2	2	1.7	1	1	1	1.0

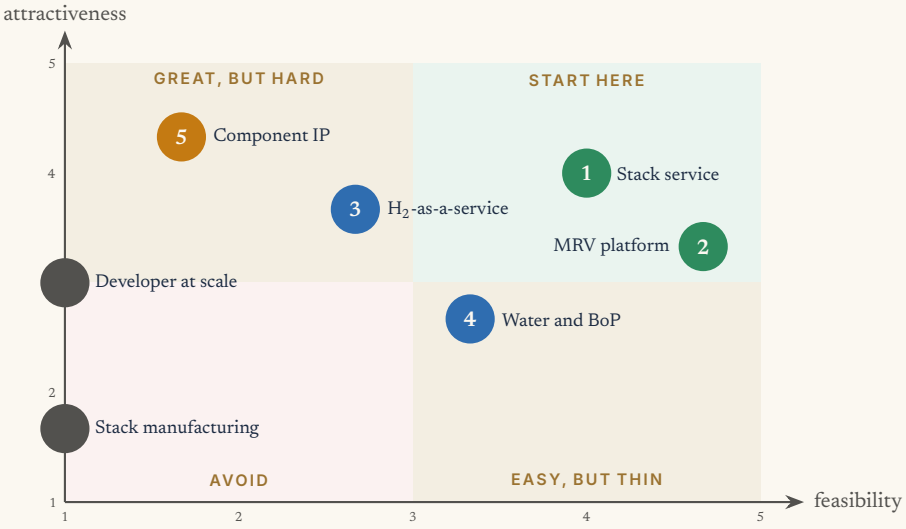


Figure 5. *Attractiveness against feasibility. The upper-right quadrant is where I would start.*

6.2 The five, with reasoning

1. Independent stack service, testing and refurbishment. *The idea:* become the company that keeps electrolysers running regardless of who made them. That means performance testing, degradation analytics, stack refurbishment, spare parts and, eventually, performance guarantees that lenders and insurers accept. *Why it is attractive:* every one of the 862,000 t/yr of allocated plants [?] will need servicing for 25 years, and the OEMs are fragile. Cummins has stopped selling electrolysers [?]; Enapter is absorbing returned stacks [?]; some Indian makers hold only 100 MW a year of incentive capacity [?]. Lenders will want somebody independent to stand behind a plant if its OEM disappears. *Why it is feasible:* it needs workshops and engineers, not gigafactories, and the analytics core (degradation curves, failure prediction) is exactly the Python and machine-learning work we are already planning for GitHub. *The risk:* few plants are operating yet (about 8,000 t/yr [?]), so revenue arrives as the fleet is built. Start with testing and analytics while waiting. MEDIUM

2. Certification and hourly MRV platform. *The idea:* software and verification support that proves, hour by hour, that a plant's hydrogen meets India's 2 kg standard [?] *and* the EU's RFNBO rules, so one data trail satisfies BEE's accredited verifiers [?], European buyers [?] and the EU border carbon charge [?]. *Why it is attractive:* certification is mandatory, recurring and data-heavy, and exporters face two rulebooks at once. *Why it is feasible:* the cheapest to start (₹5–50 crore ESTIMATED) and the closest fit to a finance-and-analytics team. *The risk:* it is the easiest to copy, and large certifiers could build it themselves. Its defensibility comes from bundling it with entry point 1, where the operating data originates. MEDIUM

3. Hydrogen-as-a-service for mid-sized factories. *The idea:* build, own and operate small on-site plants (about 0.5–5 MW) for steel, glass, food, pharma and electronics users, and sell hydrogen per kilogram on a long contract. Hygenco's plant for Jindal Stainless is the Indian proof [?]. *Why it is attractive:* the industrial-gas model earns about 30% operating margins at scale [?], contracts are long, and customers stay. *Why it is less feasible today:* each site needs ₹10–100 crore of project money ESTIMATED, and small-ticket green hydrogen finance barely exists (Section 5, gap 7). *The path:* enter once entry points 1 and 2 have produced the operating record that makes small plants financeable. MEDIUM

4. Electrolyser-ready water and balance-of-plant packages. *The idea:* standard, pre-engineered skids for desalination, purification and compression, sized for 5–100 MW electrolysers at coastal hubs [?]. *Why it ranks fourth:* real and fast to sell, but margins are thin (8–15% ESTIMATED) and established water and compression firms can enter easily. Better as a partnership than a company. MEDIUM

5. Component IP: electrodes, catalysts, iridium recycling. *The idea:* own a piece of patented technology that every stack maker needs, as Newtrace is attempting with alkaline electrodes [?], or recover iridium from spent PEM stacks, where 90%+ recycling is needed if PEM is to scale [?]. *Why it is attractive:* the highest margins and the strongest defence; patent holders attract most hydrogen venture money [?]. *Why it ranks last:* it needs scientists, labs and seven to ten years. It is an option to hold through a university partnership, not a first business. MEDIUM

6.3 From start-up to scale: the staircase

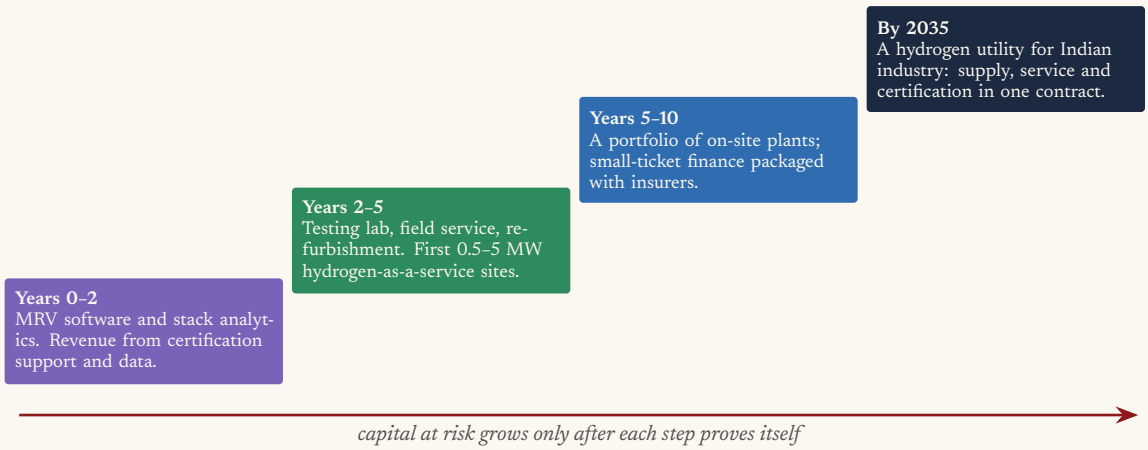


Figure 6. *The staircase from a capital-light start to a durable business. This is my plan, so every step is* PROJECTED.

IF I HAD TO START ON MONDAY

I would combine entry points 1 and 2 into one company: **the independent operator's toolkit for Indian electrolysers**, covering testing, degradation analytics, service and green certification. It needs the least capital, uses our strongest skills, sits in the high-margin, low-capital corner of the smile curve, and it grows with every plant anyone builds, whoever wins the auctions and whichever stack they choose. Hydrogen-as-a-service comes next, once our data has made small plants bankable. I would stay out of stack manufacturing and developer-scale plants until we had earned the balance sheet to survive their risks. MEDIUM

WHAT WOULD CHANGE MY MIND

Three signposts would move this ranking. First, if purchase mandates for refineries and fertiliser plants are notified [?], demand certainty jumps and developer economics improve. Second, if OEMs start bundling 20-year full-service guarantees as standard, the independent service gap narrows. Third, if AEM stacks show 20,000+ hours of clean field data, component IP in AEM becomes much more attractive.

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APA style. Every bracketed number jumps here; every entry links to the original page. Accessed October 2026.

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